

CSC343 Assignment 3 - Part 2 Solutions

1) Relation $R_1(L, M, N, O, P, Q, R, S)$, $S_1 = \{ L \rightarrow NQ, MNR \rightarrow O, O \rightarrow M, NQ \rightarrow LS, S \rightarrow OPR \}$

(a) BCNF violations

- L is a superkey since $L^+ = \text{all attributes}$.
- NQ is a superkey since $NQ^+ = \text{all attributes}$.
- O is not a superkey ($O^+ = OM$).
- S is not a superkey ($S^+ = SOPR$).
- MNR is not a superkey ($MNR^+ = MNR O$).

Violations: $MNR \rightarrow O$, $O \rightarrow M$, $S \rightarrow OPR$.

(b) BCNF decomposition

Start with $R_1(LMNOPQRS)$.

Use $O \rightarrow M$:

- $R_{11}(OM)$
- $R_{12}(LNOPQRS)$

In R_{12} , $S \rightarrow OPR$ still violates BCNF. Decompose:

- $R_{121}(OPRS)$
- $R_{122}(LNQS)$

Final BCNF relations (alphabetical):

- $LNQS$
- MO
- $OPRS$

Projected FDs:

- On $LNQS$: $L \rightarrow NQ$, $NQ \rightarrow LS$
- On MO : $O \rightarrow M$
- On $OPRS$: $S \rightarrow OPR$

(c) Dependency preservation

Not dependency-preserving.

$MNR \rightarrow O$ is not present in any single final relation and is not derivable from projected dependencies over $LNQS$, MO , and $OPRS$.

(d) Chase test (lossless)

Rows for MO , $OPRS$, $LNQS$ are initialized in the chase tableau.

Because intersection attributes determine one side at each decomposition step:

- In $R_1 \rightarrow MO$ and $LNOPQRS$, intersection is O , and $O \rightarrow MO$ holds.
- In $LNOPQRS \rightarrow OPRS$ and $LNQS$, intersection is S , and $S \rightarrow OPRS$ holds.

Applying $S \rightarrow OPR$ and then $O \rightarrow M$ to the $LNQS$ row eventually gives distinguished symbols for all attributes. Therefore the decomposition is lossless.

2) Relation $R_2(A, B, C, D, E, F, G, H)$,

$S_2 = \{ AB \rightarrow C, C \rightarrow ABD, CF \rightarrow E, E \rightarrow B, BF \rightarrow EC, B \rightarrow DA \}$

(a) Minimal basis

After splitting and minimizing:

- $AB \rightarrow C$ becomes $B \rightarrow C$ (A is extraneous because $B \rightarrow A$).
- $C \rightarrow A$ and $C \rightarrow D$ are redundant ($C \rightarrow B$ and $B \rightarrow A, D$).
- $BF \rightarrow C$ is redundant ($B \rightarrow C$).
- $CF \rightarrow E$ is redundant ($C \rightarrow B$ and $BF \rightarrow E$).

A minimal basis (alphabetical) is:

- $B \rightarrow A$
- $B \rightarrow C$
- $B \rightarrow D$
- $BF \rightarrow E$

- $C \rightarrow B$
- $E \rightarrow B$

(b) All keys for R2

Attributes not on RHS are F,G,H, so every key must include FGH.

Adding one of {B, C, E} yields all attributes:

- BFGH
- CFGH
- EFGH

These are all candidate keys.

(c) 3NF synthesis

Combine same LHS:

- $B \rightarrow ACD$
- $BF \rightarrow E$
- $C \rightarrow B$
- $E \rightarrow B$

Create relations:

- R1(ABCD)
- R2(BEF)
- R3(BC)
- R4(BE)

Remove subset relations:

- BC is subset of ABCD, remove BC.
- BE is subset of BEF, remove BE.

No relation contains a key for R2, so add one key relation, e.g. BFGH.

Final 3NF decomposition:

- ABCD
- BEF
- BFGH

(d) Redundancy

Yes, redundancy can still exist in this 3NF design.

In relation BEF, projected dependency $E \rightarrow B$ has non-superkey LHS E.

So repeated B values can occur for the same E across multiple F values.